Amir Mirzai Golpayegani

Calgary, AB, Canada

amirgolpaa24@gmail.com | amir.mirzaigolpayega@ucalgary.ca | +1 (587) 439-0436

linkedin.com/in/amir-golpayegani-218148165 | github.com/amirgolpaa24 | Portfolio

Summary

M.Sc. Computer Science graduate with experience developing, evaluating, and validating machine learning and deep neural network models in Python and PyTorch. Skilled in model evaluation, custom metrics, reproducible experiments, data preprocessing, SQL-backed backend systems, and research communication, with applied experience analyzing generalization, documenting tradeoffs, and building AI workflow automation.

Education

M.Sc. in Computer Science — GPA: **3.94**/4.0 — Sep 2022 – Feb 2026
University of Calgary — *Calgary, Canada*

B.Sc. in Computer Engineering — GPA: **3.69**/4.0 — Sep 2016 – Feb 2021
Sharif University of Technology — *Tehran, Iran*

Technical Skills

Programming: Python, C++, JavaScript, SQL
ML / AI: PyTorch, Scikit-learn, TensorFlow, CNNs, model training, validation, evaluation, LLM workflows
Data / Evaluation: NumPy, Pandas, SciPy, custom metrics, regression-style checks, data preprocessing, drift-aware analysis
Engineering Tools: Linux, Git, Docker, Django, Node.js, REST APIs, PostgreSQL, AWS, GitHub Actions
Visualization / Automation: Streamlit, Polyscope, OpenGL, GLSL, n8n, OpenClaw, SearXNG, Telegram API

Work Experience

Research Software Developer — Sep 2022 – Feb 2026
BigGeo / Mitacs, University of Calgary — *Calgary, Canada*

- Built C++/Python software for scalable processing, storage, and retrieval of large Earth observation datasets.
- Automated SAR and optical imagery preprocessing using GDAL, Rasterio, SNAP, and satellite data APIs.
- Developed DGGS-based encoding pipelines to convert satellite imagery into structured tensor files for analysis.
- Improved encoding performance with C++ multithreading, reducing 10-tensor processing from 233s to 26s.
- Reduced storage by 38% for scaled Canada-wide data with 1,112 GeoTIFF tiles and 16K+ DGGS tensors.

Deep Learning Teaching Assistant — Jul 2026
Neuromatch Academy — *Remote, United States*

- Mentored an international pod of $\sim$15 students through Neuromatch Academy's intensive three-week Deep Learning course.
- Facilitated hands-on PyTorch tutorials spanning MLPs, CNNs, attention/transformers, diffusion generative models, and reinforcement learning.
- Guided project groups through end-to-end deep learning research projects, from question formulation and literature review to model implementation and final presentations.

Back-End Developer — Jun 2020 – Jul 2022
Asr Gooyesh Pardaz — *Tehran, Iran*

- Developed Django REST APIs for NLP, text-to-speech, and speech-to-text service plans backed by PostgreSQL.
- Built authentication features including registration, Google login, password reset, and email/phone verification.
- Implemented backend logic for payment-service integration and storing plan/payment records in a relational database.
- Contributed to a WebRTC-based chatbot system using GPT-style responses and FAQ data to answer client questions.

Teaching Assistant — Dec 2022 – Apr 2025
Department of Computer Science, University of Calgary — *Calgary, Canada*

- **Developing Big Data Applications**: Mentored +25 students on AWS-based data pipelines, cloud storage, debugging, and scalable data workflows.
- **Working with Data at Scale**: Supported +20 students with large-scale data processing, data pipelines, and high-volume data analysis.
- **Database Management Systems**: Assisted +75 students with SQL, database design, course projects, grading, and exam support.

Research and Technical Projects

DEM Super-Resolution Framework — May 2024 – Feb 2026
Python, PyTorch, CNNs, Scientific Imaging, Google Earth Engine API, Polyscope, Data Processing and Visualization

- Trained deep ResNet models for 4$\times$ DEM super-resolution, reconstructing 30m elevation data from 120m inputs.
- Built a Streamlit pipeline using Rasterio to parallelize GEE DEM tile downloads and collect mountainous terrain data

worldwide.
- Implemented PyTorch training and evaluation workflows with terrain-aware metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors.
- Analyzed model generalization across geographic regions and used evaluation results to compare reconstruction quality.

Agentic Job Search Automation Platform May 2026 – Present
OpenClaw, n8n, LLMs, AI Agents, Docker, Workflow Automation, SearXNG, Telegram API
- Building an agentic AI system to search jobs, analyze job fit, and generate tailored application materials.
- Designed LLM-based workflows for resume/job matching, skill-gap analysis, and application decision support.
- Integrated Google Sheets, Google Drive, and Telegram API to track jobs and send automated application alerts.
- Prototyped the initial workflow in n8n before migrating toward a more flexible OpenClaw-based agent system.

Spatiotemporal Earth Observation Data System Using DGGS Sep 2022 – Feb 2026
First-author publication | *M.Sc. thesis* | *C++, Python, DGGS, Wavelets, B-splines, Time-Series*
- Designed a DGGS-based framework for real-time multiresolution management of satellite imagery.
- Developed a tensor-based storage model for direct access to arbitrary regions at multiple resolutions.
- Retrieved 105,489 DGGS cells per time-series query in 13.9s, compared with about 220s using global retrieval.

Publication

Amir Mirzai Golpayegani, Mahmudul Hasan, Faramarz F. Samavati. *Real-Time Multiresolution Management of Spatiotemporal Earth Observation Data Using DGGS*. Remote Sensing, 2025. doi:10.3390/rs17040570

Certificates

Supervised Machine Learning: Regression and Classification, DeepLearning.AI and Stanford University, Coursera Oct 2024
Applied Machine Learning in Python, University of Michigan, Coursera Jul 2020

Languages

English: Full Professional Proficiency (TOEFL **115**/120)
Persian: Native Proficiency

Achievement

Teaching Assistant Excellence Laureate 2024/2025
Department of Computer Science, University of Calgary — awarded for outstanding teaching support (*link*)

Ranked 17th Nationwide in Iran University Entrance Exam (Konkour) May 2016
Top 0.001% among all participants